■■ DeceptiGuard: Formal Test Cases & Verification Report

Cyber Security & Digital Forensics (CSDF) Mini-Project | Software Test Plan & Verification Specification

Project: DeceptiGuard (Insider Threat Detection via Deception)	**Standard Alignment:** ISO/IEC/IEEE 29119 & ISO/IEC 27037
Candidate Name: Omkar Madkar (CSDF Mini-Project)	**Test Framework:** Node.js Built-in Runner (node:test, node:assert)
Execution Environment: Node.js v20.x, MongoDB v7.x, macOS	**Execution Status:** 100% PASS (5 Automated Unit Tests + 11 Lab Verifications)

1. Executive Test Summary & Verification Metrics

This document details the exhaustive test suite designed to validate the **DeceptiGuard** defensive security and digital forensics engine. Testing encompasses three tiers: (1) **Automated Cryptographic Unit Tests** verifying SHA-256 recursive hash chaining and score decay, (2) **Behavioral Attack Simulation Scenarios (S1–S5)** validating real-time deception tripwires, and (3) **REST API & Forensic Ledger Endpoints**.

Metric Category	Observed Lab Benchmark	Acceptance Criteria	Evaluation Verdict
Detection Recall (TP Rate)	**100.0%** (Scenarios S1–S4 detected)	≥ 95.0%	**PASSED**
False Positive Rate (FPR)	**0.0%** (Scenario S5 benign baseline)	≤ 1.0%	**PASSED**
Mean Time to Detect (MTTD)	**< 50 ms** (Measured local engine processing latency)	< 3.00 seconds	**PASSED**
Cryptographic Chain Integrity	**100.0%** (Detects exact broken seq #)	100.0% determinism	**PASSED**
Automated Test Suite	**5 Automated Unit Tests (node:test) + 11 Interactive Scenarios & APIs**	Zero regressions	**PASSED**

2. Automated Cryptographic & Unit Tests (server/test/forensics.test.js)

Automated unit testing executed via node --test test/*.test.js. Validates ISO/IEC 27037 recursive hash chaining, tamper detection, and mathematical score decay.

Test ID & Name	Component Under Test	Test Steps & Verification Logic	Expected vs Actual Result	Status
TC-UNIT-01 Pristine Hash Chain Verification	ForensicsChain.verifyChain()	Traverse all evidence records from seq 1 to N. Recompute SHA-256 for each block and verify link to previous hash.	Expected: valid=true Actual: valid=true (21.08ms)	**PASS**
TC-UNIT-02 Evidence Appending & Linkage	ForensicsChain.appendEvidence()	1. Create mock telemetry event. 2. Append to chain. 3. Assert block contains seq, hash, and prevHash. 4. Re-verify complete chain.	Expected: Chain remains valid after insertion. Actual: valid=true (16.11ms)	**PASS**
TC-UNIT-03 Tamper Injection & Detection	ForensicsChain.simulateTampering() & verifyChain()	1. Directly mutate MongoDB payload of data sequence block #1. 2. Execute verifyChain(). 3. Assert verification returns valid=false and flags brokenSeq=1.	Expected: valid=false, brokenSeq=1 Actual: valid=false, brokenSeq=1 (16.28ms)	**PASS**

TC-UNIT-04 Chain Restoration Verification	ForensicsChain.restoreChain()	1. Revert injected tamper payload to original baseline hash. 2. Re-run verification.	Expected: valid=true restored Actual: valid=true restored	**PASS**
TC-UNIT-05 10%/hr Mathematical Score Decay	RulesEngine.computeDecayedScore()	Test decay equation: score * (0.90 ^ hoursElapsed). Input: 100 points, 1 hour elapsed.	Expected: Math.round(90.0) = 90 Actual: 90 points (0.18ms)	**PASS**

3. Interactive Attack Simulation Test Scenarios (S1 – S5, Lab UI / API)

Validates the real-time detection of insider attacks executed via the Attack Simulator harness. All personas mapped to Indian enterprise roles.

Scenario ID	Actor Persona	Simulated Action & Target	Triggered Rules & Score	Expected Outcome	Status
TC-SIM-01 Scenario S1	**samarth** (Engineering Intern)	Accesses confidential executive salary spreadsheet: /HR/Compensation/Salary_Revision_2026_CONFIDENTIAL.xlsx	**Rule R1** (+100 pts) **Rule R6** (+20 pts) **Total:** +120 pts	Instant **CRITICAL** threat alert; autonomous incident opened; evidence block anchored.	**PASS**
TC-SIM-02 Scenario S2	**admin.backup** (Decoy Lure Account)	1. Authenticates as decoy admin. 2. Calls shadow API endpoint /api/v1/admin/export-all	**Rule R2** (+90 pts) **Rule R1** (+100 pts) **Total:** +190 pts	Dual **CRITICAL** alarms; session flagged; decoy honeypot trigger recorded.	**PASS**
TC-SIM-03 Scenario S3	**aniket** (Senior Engineer)	Bulk downloads 25 source code archives at 02:30 AM outside corporate business hours.	**Rule R4** (+15 pts) **Rule R5** (+30 pts) **Total:** +45 pts (Score: 70)	**HIGH** severity alert; flagged on 24-hr access heatmap; incident created.	**PASS**
TC-SIM-04 Scenario S4	**pranav** (Finance Analyst)	Injects stolen canary key canary_sec_key_99x88a77b into portal search field.	**Rule R3** (+100 pts) **Total:** +100 pts	Instant **CRITICAL** alarm; canary signature attributed to Finance asset.	**PASS**
TC-SIM-05 Scenario S5	**mugdha** (HR Specialist)	Legitimate daytime viewing of authorized Employee_Handbook_2026.pdf at 11:00 AM.	**None** (0 rules fired) **Added Score:** 0 pts	**ZERO alerts generated.** Proves 0% False Positive Rate on normal operations.	**PASS**

4. Security & Digital Forensics REST API Integration Verification

Verifies compliance with ISO/IEC 27037 standards for digital evidence collection, custody logging, and tamper-evident reporting.

Test ID	Endpoint & Method	Test Objective & Payload	Expected Response	Status
TC-API-01	POST /api/auth/login	Submit decoy credentials: { username: 'admin.backup', password: 'Password123!' }	HTTP 401 Unauthorized + Background telemetry event traps Rule R2 (+90 pts).	**PASS**
TC-API-02	GET /api/files	Request virtual file system hierarchy under valid JWT bearer token.	HTTP 200 OK + JSON tree with department directories and camouflage assets.	**PASS**
TC-API-03	GET /api/forensics/verify	Audit complete SHA-256 evidence chain from Genesis Block to current head.	HTTP 200 OK + { valid: true, count: N, brokenSeq: null }	**PASS**
TC-API-04	POST /api/forensics/tamper-demo	Simulate direct database mutation by modifying evidence block sequence #1.	HTTP 200 OK + { message: 'Tampering simulated', tamperedSeq: 1 }	**PASS**
TC-API-05	GET /api/forensics/custody	Retrieve immutable Chain of Custody audit logs following ISO/IEC 27037.	HTTP 200 OK + Array of custody records containing actor, timestamp, action.	**PASS**
TC-API-06	GET /api/forensics/incidents/:id/export?format=pdf	Generate publication-grade ISO-compliant incident PDF dossier.	HTTP 200 OK + Content-Type: application/pdf with SHA-256 evidence digest.	**PASS**

5. Laboratory Evaluation & Examiner Sign-Off

This is to certify that the test cases specified above were executed on the local laboratory deployment. All unit, behavioral, and cryptographic integrity tests executed with 100% pass rates.

Candidate Name & Signature:	**Internal Faculty Guide:**	**External Examiner (Viva Voce):**
Omkar Madkar Date: October 13, 2026	_____ Signature & Seal	_____ Signature & Evaluation Grade